

Tutorial 7 (Cam)

Ignore plotting

- Problem 8-1
- Problem 8-4
- Problem 8-46
- Problem 8-9
- Problem 8-8

*8-1 Figure P8-1 shows the cam and follower from Problem 6-65. Using graphical methods, find and sketch the equivalent fourbar linkage for this position of the cam and follower.

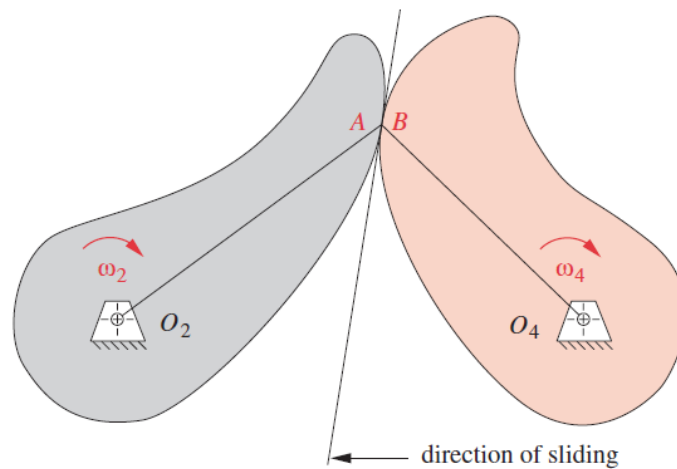


FIGURE P8-1

*8-4 Figure P8-2 shows a cam and follower. Using graphical methods, find the pressure angle at the position shown.

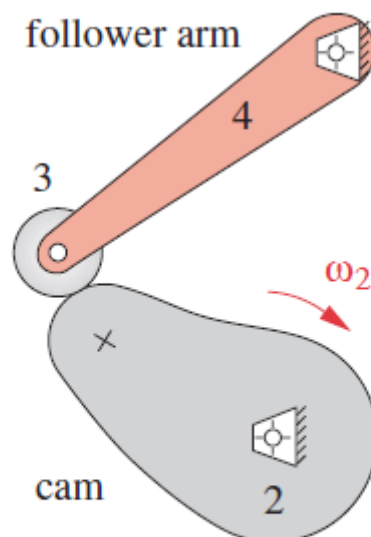


FIGURE P8-2

- ‡8-46 Design a double-dwell cam to move a follower from 0 to 50 mm in 75° , dwell for 75° , fall 50 mm in 75° , and dwell for the remainder. The total cycle must take 5 sec. Use a 4-5-6-7 polynomial function for rise and fall and plot the $s \ v \ a \ j$ diagrams.
- ‡8-9 Design a single-dwell cam to move a follower from 0 to 2" in 60° , fall 2" in 90° , and dwell for the remainder. The total cycle must take 2 sec. Choose suitable functions for rise and fall to minimize accelerations. Plot the $s \ v \ a \ j$ diagrams.
- ‡8-8 Design a double-dwell cam to move a follower from 0 to 1.5" in 45° , dwell for 150° , fall 1.5" in 90° , and dwell for the remainder. The total cycle must take 6 sec. Choose suitable functions for rise and fall to minimize velocities. Plot the $s \ v \ a \ j$ diagrams.